

# **MLCC 2026**

## **Regularization and linear models**

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## About this class

- ▶ We introduce a class of learning algorithms based on *Tikhonov regularization*
- ▶ We study computational aspects of these algorithms .

## Empirical Risk Minimization (ERM)

- ▶ Empirical Risk Minimization (ERM): probably the most popular approach to design learning algorithms.
- ▶ **General idea:** considering the empirical error

$$\hat{\mathcal{E}}(f) = \frac{1}{n} \sum_{i=1}^n \ell(y_i, f(x_i)),$$

as a proxy for the expected error

$$\mathcal{E}(f) = \mathbb{E}[\ell(y, f(x))] = \int dx dy p(x, y) \ell(y, f(x)).$$

# The Expected Risk is Not Computable

Recall that

- ▶  $\ell$  measures the price we pay predicting  $f(x)$  when the true label is  $y$
- ▶  $\mathcal{E}(f)$  cannot be directly computed, since  $p(x, y)$  is unknown

## From Theory to Algorithms: The Hypothesis Space

To turn the above idea into an actual algorithm, we:

- ▶ Fix a suitable hypothesis space  $H$
- ▶ Minimize  $\hat{\mathcal{E}}$  over  $H$

$H$  should allow feasible computations and be *rich*, since the complexity of the problem is not known a priori.

## Example: Space of Linear Functions

The simplest example of  $H$  is the space of linear functions:

$$H = \{f : \mathbb{R}^d \rightarrow \mathbb{R} : \exists w \in \mathbb{R}^d \text{ such that } f(x) = x^T w, \forall x \in \mathbb{R}^d\}.$$

- ▶ Each function  $f$  is defined by a vector  $w$
- ▶  $f_w(x) = x^T w$ .

## Rich Hypothesis spaces May Require Regularization

- ▶ If  $H$  is rich enough, solving ERM may cause overfitting (solutions highly dependent on the data)
- ▶ Regularization induces a bias in the search of a solution ensuring stability and generalization

# Tikhonov Regularization

Consider the *Tikhonov* regularization scheme,

$$\min_{w \in \mathbb{R}^d} \hat{\mathcal{E}}(f_w) + \lambda \|w\|^2 \quad (1)$$

It describes a large class of methods sometimes called Regularization Networks.



# The Regularizer

- ▶  $\|w\|^2$  is called *regularizer*
- ▶ It controls the stability of the solution and prevents overfitting
- ▶  $\lambda$  balances the error term and the regularizer

# Loss Functions

- ▶ Different loss functions  $\ell$  induce different classes of methods
- ▶ We will see common aspects and differences in considering different loss functions
- ▶ There exists no general computational scheme to solve Tikhonov Regularization
- ▶ The solution depends on the considered loss function

# The Regularized Least Squares Algorithm

## Regularized Least Squares: *Tikhonov* regularization

$$\min_{w \in \mathbb{R}^d} \hat{\mathcal{E}}(f_w) + \lambda \|w\|^2, \quad \hat{\mathcal{E}}(f_w) = \frac{1}{n} \sum_{i=1}^n \ell(y_i, f_w(x_i)) \quad (2)$$

Square loss function:

$$\ell(y, f_w(x)) = (y - f_w(x))^2$$

We then obtain the RLS optimization problem (linear model):

$$\min_{w \in \mathbb{R}^d} \frac{1}{n} \sum_{i=1}^n (y_i - w^T x_i)^2 + \lambda w^T w, \quad \lambda \geq 0. \quad (3)$$

## Matrix Notation

- ▶ The  $n \times d$  matrix  $\hat{X}$ , whose rows are the input points
- ▶ The  $n \times 1$  vector  $\hat{Y}$ , whose entries are the corresponding outputs.

With this notation,

$$\frac{1}{n} \sum_{i=1}^n (y_i - w^T x_i)^2 = \frac{1}{n} \|\hat{Y} - \hat{X}w\|^2.$$

## Gradients of the ER and of the Regularizer

By direct computation,

- Gradient of the empirical risk w. r. t.  $w$

$$-\frac{2}{n}\hat{X}^T(\hat{Y} - \hat{X}w)$$

- Gradient of the regularizer w. r. t.  $w$

$$2w$$

## The RLS Solution

By setting the gradient to zero, the solution of RLS solves the linear system

$$(\hat{X}^T \hat{X} + \lambda n I) w = \hat{X}^T \hat{Y}.$$

from which we derive

$$w^* = \frac{\hat{X}^T \hat{Y}}{\hat{X}^T \hat{X} + \lambda n I}$$

Hence  $\lambda$  controls the *invertibility* of  $(\hat{X}^T \hat{X} + \lambda n I)$

## Choosing the Cholesky Solver

- ▶ Several methods can be used to solve the above linear system
- ▶ Cholesky decomposition is the method of choice, since

$$\hat{X}^T \hat{X} + \lambda n I$$

is symmetric and positive definite.

# Time Complexity

Time complexity of the method :

- ▶ Training:  $O(nd^2)$  (assuming  $n \gg d$ )
- ▶ Testing:  $O(d)$



## Dealing with an Offset

For linear models, especially in low dimensional spaces, it is useful to consider an *offset*:

$$w^T x + b$$

How to estimate  $b$  from data?

## Idea: Augmenting the Dimension of the Input Space

- ▶ Simple idea: augment the dimension of the input space, considering  $\tilde{x} = (x, 1)$  and  $\tilde{w} = (w, b)$ .
- ▶ This is fine if we do not regularize, but if we do then this method tends to prefer linear functions passing through the origin (zero offset), since the regularizer becomes:

$$\|\tilde{w}\|^2 = \|w\|^2 + b^2.$$

## Avoiding to Penalize the Solutions with Offset

We want to regularize considering only  $\|w\|^2$ , without penalizing the offset.

The modified regularized problem becomes:

$$\min_{(w,b) \in \mathbb{R}^{d+1}} \frac{1}{n} \sum_{i=1}^n (y_i - w^T x_i - b)^2 + \lambda \|w\|^2.$$

## Solution with Offset: Centering the Data

It can be proved that a solution  $w^*, b^*$  of the above problem is given by

$$b^* = \bar{y} - \bar{x}^T w^*$$

where

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

## Solution with Offset: Centering the Data

$w^*$  solves

$$\min_{w \in \mathbb{R}^d} \frac{1}{n} \sum_{i=1}^n (y_i^c - w^T x_i^c)^2 + \lambda \|w\|^2.$$

where  $y_i^c = y_i - \bar{y}$  and  $x_i^c = x_i - \bar{x}$  for all  $i = 1, \dots, n$ .

**Note:** This corresponds to centering the data and then applying the standard RLS algorithm.

# Introducing: Regularized Logistic Regression

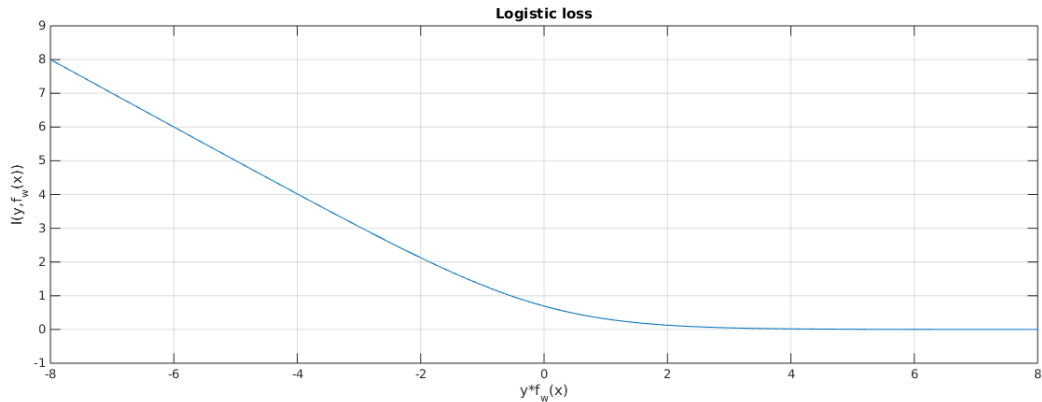
**Regularized logistic regression:** *Tikhonov* regularization

$$\min_{w \in \mathbb{R}^d} \hat{\mathcal{E}}(f_w) + \lambda \|w\|^2, \quad \hat{\mathcal{E}}(f_w) = \frac{1}{n} \sum_{i=1}^n \ell(y_i, f_w(x_i)) \quad (4)$$

With the *logistic loss function*:

$$\ell(y, f_w(x)) = \log(1 + e^{-y f_w(x)})$$

# The Logistic Loss Function



**Figure:** Plot of the logistic regression loss function

## Minimization Through Gradient Descent

- ▶ The logistic loss function is differentiable
- ▶ The candidate to compute a minimizer is the *gradient descent (GD)* algorithm



## Regularized Logistic Regression (RLR)

- ▶ The regularized ERM problem associated with the logistic loss is called *regularized logistic regression*
- ▶ Its solution can be computed via gradient descent
- ▶ Note:

$$\nabla \hat{\mathcal{E}}(f_w) = \frac{1}{n} \sum_{i=1}^n x_i \frac{-y_i e^{-y_i x_i^T w_{t-1}}}{1 + e^{-y_i x_i^T w_{t-1}}} = \frac{1}{n} \sum_{i=1}^n x_i \frac{-y_i}{1 + e^{y_i x_i^T w_{t-1}}}$$

## RLR: Gradient Descent Iteration

For  $w_0 = 0$ , the GD iteration applied to

$$\min_{w \in \mathbb{R}^d} \hat{\mathcal{E}}(f_w) + \lambda \|w\|^2$$

is

$$w_t = w_{t-1} - \gamma \underbrace{\left( \frac{1}{n} \sum_{i=1}^n x_i \frac{-y_i}{1 + e^{y_i x_i^T w_{t-1}}} + 2\lambda w_{t-1} \right)}_a$$

for  $t = 1, \dots, T$ , where

$$a = \nabla(\hat{\mathcal{E}}(f_w) + \lambda \|w\|^2)$$

## Logistic Regression and Confidence Estimation

- ▶ The solution of logistic regression has a probabilistic interpretation
- ▶ It can be derived from the following model

$$p(1|x) = \frac{e^{x^T w}}{\underbrace{1 + e^{x^T w}}_{h(w)}}$$

where  $h$  is called *logistic function*.

- ▶ This can be used to compute a *confidence* for each prediction

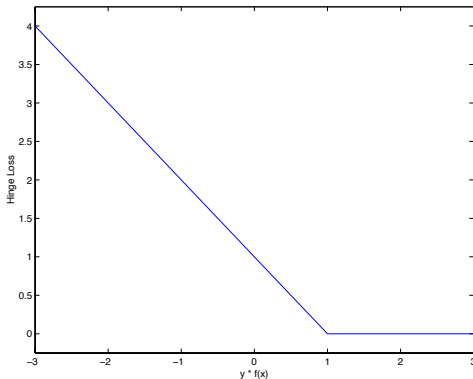
# Support Vector Machines

**Formulation in terms of Tikhonov regularization:**

$$\min_{w \in \mathbb{R}^d} \hat{\mathcal{E}}(f_w) + \lambda \|w\|^2, \quad \hat{\mathcal{E}}(f_w) = \frac{1}{n} \sum_{i=1}^n \ell(y_i, f_w(x_i)) \quad (5)$$

With the *Hinge loss function*:

$$\ell(y, f_w(x)) = |1 - y f_w(x)|_+$$



## A more classical formulation (linear case)

$$w^* = \arg \min_{w \in \mathbb{R}^d} \frac{1}{n} \sum_{i=1}^n |1 - y_i w^\top x_i|_+ + \lambda \|w\|^2$$

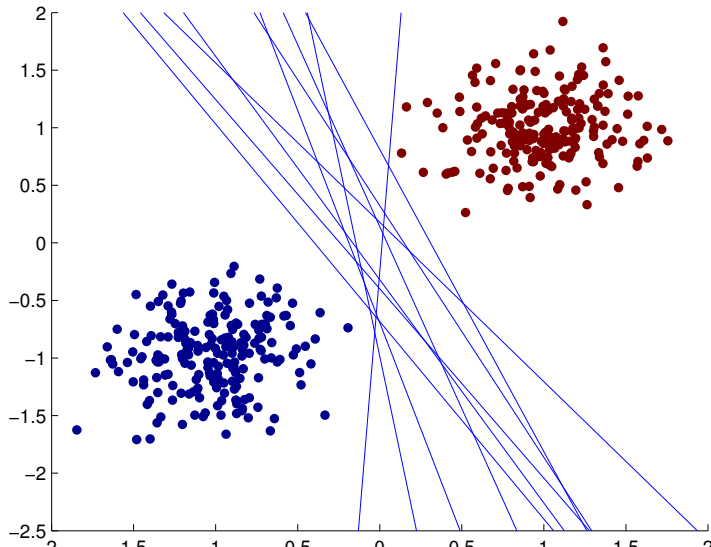
with  $\lambda = \frac{1}{C}$

## A more classical formulation (linear case)

$$w^* = \arg \min_{w \in \mathbb{R}^d, \xi_i \geq 0} \|w\|^2 + \frac{C}{n} \sum_{i=1}^n \xi_i \quad \text{subject to}$$
$$y_i w^\top x_i \geq 1 - \xi_i \quad \forall i \in \{1 \dots n\}$$

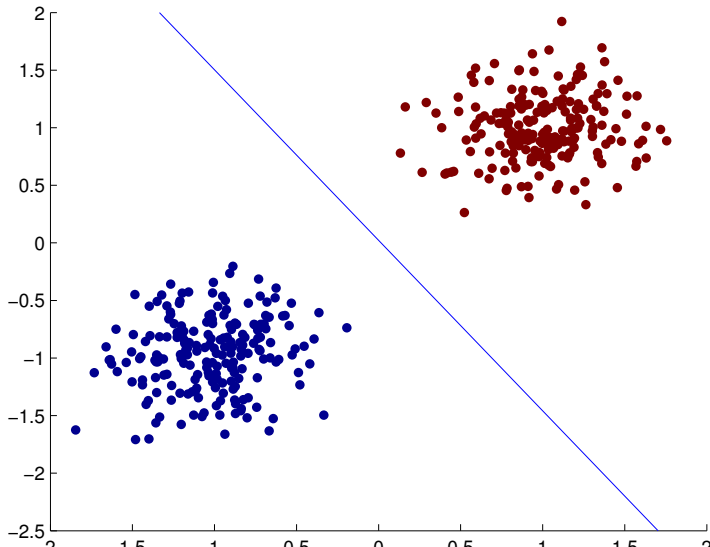
## A geometric intuition - classification

In general do you have many solutions



## A geometric intuition - classification

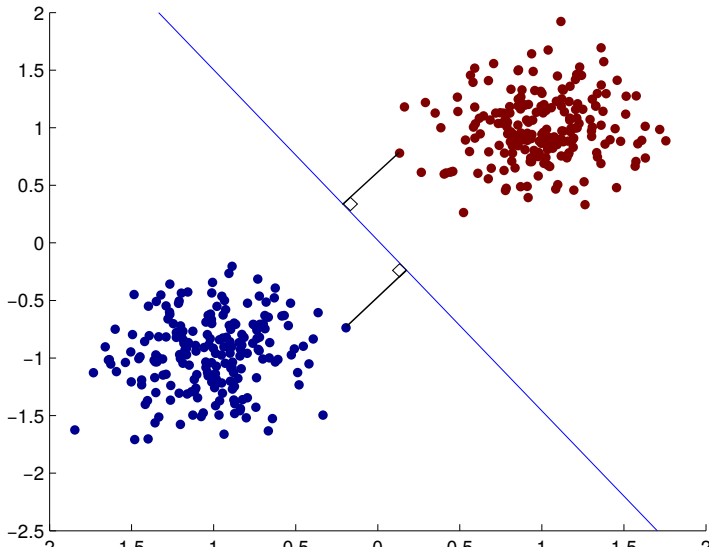
Intuitively I would choose an “equidistant” line





## A geometric intuition - classification

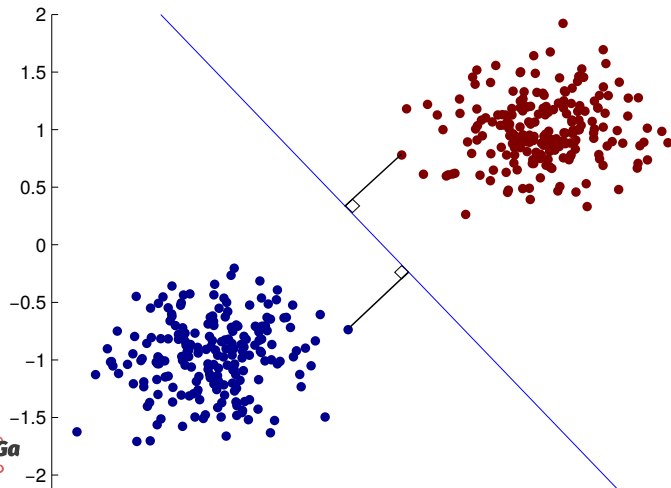
Intuitively I would choose an “equidistant” line



## Maximum margin classifier

I want the classifier that

- ▶ classifies perfectly the dataset
- ▶ maximize the distance from its closest examples



## Margin

An hyperplane  $w$  well classifies an example  $(x_i, y_i)$  if

- ▶  $y_i = 1$  and  $w^\top x_i > 0$  or
- ▶  $y_i = -1$  and  $w^\top x_i < 0$

therefore  $x_i$  is well classified iff  $y_i w^\top x_i > 0$

**Margin:**  $m_i = y_i w^\top x_i$

Note that  $x_\perp = x - \frac{y_i m_i}{\|w\|} w$

## Maximum margin classifier definition

I want the classifier that

- ▶ classifies perfectly the dataset
- ▶ maximize the distance from its closest examples

$$w^* = \max_{w \in \mathbb{R}^d} \min_{1 \leq i \leq n} d(x_i, w)^2 \quad \text{subject to}$$
$$m_i > 0 \quad \forall i \in \{1 \dots n\}$$

Let call  $\mu$  the smallest  $m_i$  thus we have

$$w^* = \max_{w \in \mathbb{R}^d} \min_{1 \leq i \leq n, \mu \geq 0} \|x_i\| - \frac{(x_i^\top w)^2}{\|w\|^2} \quad \text{subject to}$$
$$y_i w^\top x_i \geq \mu \quad \forall i \in \{1 \dots n\}$$

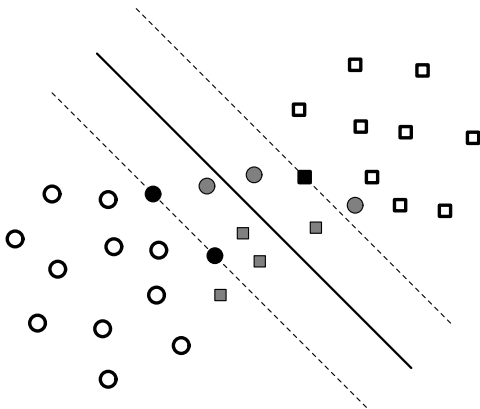
that is

## What if the problem is not separable?

We relax the constraints and penalize the relaxation

$$w^* = \min_{w \in \mathbb{R}^d} \|w\|^2 \quad \text{subject to}$$

$$y_i w^\top x_i \geq 1 \quad \forall i \in \{1 \dots n\}$$



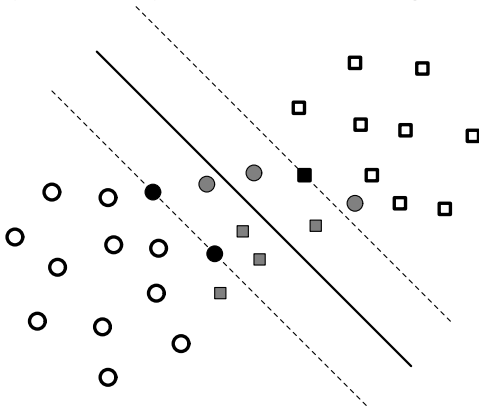
## What if the problem is not separable?

We relax the constraints and penalize the relaxation

$$w^* = \min_{w \in \mathbb{R}^d, \xi_i \geq 0} \|w\|^2 + \frac{C}{n} \sum_{i=1}^n \xi_i \quad \text{subject to}$$

$$y_i w^\top x_i \geq 1 - \xi_i \quad \forall i \in \{1 \dots n\}$$

where  $C$  is a penalization parameter for the average error  $\frac{1}{n} \sum_{i=1}^n \xi_i$ .



## Dual formulation

It can be shown that the solution of the SVM problem is of the form

$$w = \sum_{i=1}^n \alpha_i y_i x_i$$

where  $\alpha_i$  are given by the solution of the following quadratic programming problem:

$$\begin{aligned} \max_{\alpha \in \mathbb{R}^n} \quad & \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n y_i y_j \alpha_i \alpha_j x_i^T x_j \quad i = 1, \dots, n \\ \text{subj to} \quad & \alpha_i \geq 0 \end{aligned}$$

- ▶ The solution requires the estimate of  $n$  rather than  $d$  coefficients
- ▶  $\alpha_i$  are often sparse. The input points associated with non-zero coefficients are called *support vectors*

## Wrapping up

### Regularized Empirical Risk Minimization

$$w^* = \min_{w \in \mathbb{R}^d} \frac{1}{n} \sum_{i=1}^n \ell(y_i, w^\top x_i) + \lambda \|w\|^2$$

### Examples of Regularization Networks

- ▶  $\ell(y, t) = (y - t)^2$  (Square loss) leads to Least Squares
- ▶  $\ell(y, t) = \log(1 + e^{-yt})$  (Logistic loss) leads to Logistic Regression
- ▶  $\ell(y, t) = |1 - yt|_+$  (Hinge loss) leads to Maximum Margin Classifier



## Next class

... more on optimization!